Data :

x	y	$x \times y$	x^2
5	1.2	6	25
10	2.5	25	100
15	3.7	55.5	225
20	4.9	98	400
25	6.1	152.5	625
Sum Σx = 75	18.4	337	1375

$\Sigma x = 75$ $\Sigma y = 18.4$ $\Sigma xy = 337$ $\Sigma x^2 = 1375$

$n = 5$

$$m = \frac{n(\Sigma xy) - (\Sigma x \cdot \Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$m = 0.244$

$$c = \frac{\Sigma y - m(\Sigma x)}{n}$$

$c = 0.02$

$\therefore$

$$\boxed{y = 0.244x + 0.02}$$

$y = mx + b$

For Load of 55 N : $x = 55N$

Extension $= y$

$y = 0.244(55) + 0.02$

$y = 13.44$

Extension = 13.44 mm